

Do Gaussian Splats Keep the Nuclei?

An Object-Level Audit of Gaussian Volume Compression for Fluorescence Microscopy

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Abstract—Gaussian splats are being adopted to store and stream large 3D microscopy and medical volumes, and their storage budgets are chosen and reported by PSNR. We ask the question PSNR cannot answer: at a given number of bytes on disk, can standard analysis still find the nuclei? On densely annotated *C. elegans* embryos (192 and 362 labelled nuclei), we compare a Gaussian-splat compressor (Luxar) with JPEG-XL, JPEG2000, ZFP and downsampling at matched stored bytes, scoring two detectors against human labels. JPEG-XL loses no nuclei up to its maximum compression ($82\text{--}91\times$). Beyond $120\times$, splats are the best option tested (82% of nuclei kept at $\approx 250\times$ vs. 74% for JPEG2000), but how many survive depends on the detector: a blob detector keeps 98%, a learned segmenter (Cellpose) keeps 82%, and at $55\times$ to $94\times$ the segmenter already loses 8–17% of nuclei that JPEG-XL keeps. Loss collapses below ≈ 3 splats per nucleus, while Luxar’s PSNR stays within 19.1–21.3 dB across an $18\times$ range of budgets. A label-free survival check tracks the labelled result (Spearman $\rho = 0.68$) and guards against the budget cliff, though not against the segmenter’s appearance gap.

I. Introduction

Light-sheet and confocal time-lapse microscopy produce terabytes per experiment, and medical archives face the same pressure. Representing a volume as a set of anisotropic 3D Gaussians [1] is attractive for such data: the representation is compact, continuous and renders directly on the GPU. Recent systems fit Gaussian splats to microscopy volumes at scale [2], to slice-based microscopy and ultrasound [3], to medical volumes for compact storage [4], [5], and to tomographic reconstruction [6], [7]. Throughout, quality is measured by PSNR or SSIM, and budgets are chosen by PSNR: Luxar selects its splat count by self-supervised held-out PSNR [8] and explicitly describes its fits as intended for visualisation rather than quantitative analysis [2].

What a microscopist needs from a compressed volume is that the analysis still works. Whether a nucleus is still detectable is decided by a small, often faint, peak—exactly the kind of structure an intensity-weighted fitting objective can trade away. Radiology learned the same lesson for lossy codecs, where acceptability is judged by whether lesions remain detectable, not by PSNR [9]. Gaussian volume compression has not yet been held to that standard.

Contributions. (i) A rate–detectability protocol for volumetric compression: matched bytes on disk, dense human

labels, two independent detectors, one-to-one matching in physical units, and a faint/bright split. (ii) Evidence on densely labelled *C. elegans* embryos that JPEG-XL is safe up to its maximum ratio, that Gaussian splats are the best option tested at higher ratios, and that their benefit depends on the downstream detector while the codecs do not. (iii) A budget cliff near 3 splats per nucleus that PSNR does not reveal, and a label-free check that does.

II. Related Work

Gaussian mixtures have long modelled nuclei for segmentation and lineage tracking [10]; our question concerns Gaussians as a storage format. Gaussian splatting [1] has been adapted to CT [7], [6], to slice-to-volume MRI with closed-form point-spread functions [11], and to volume compression [2], [3], [4], [5], each evaluated by image-fidelity metrics. Scientific and image codecs—JPEG2000 [12], JPEG-XL [13], and the error-bounded ZFP [14]—are the natural baselines at matched size. Cell detection is benchmarked by the Cell Tracking Challenge [15], [16]; we use its labels and two standard detectors, a scale-normalised Laplacian of Gaussian [17] and Cellpose [18].

III. Methods

Data. Fluo-N3DH-CE from the Cell Tracking Challenge [16]: *C. elegans* embryos, $35\times 512\times 708$ voxels of $1.0\times 0.09\times 0.09\mu\text{m}$, 8-bit (12.1 MiB per frame). Every nucleus carries a tracking label; we use frames with 192 (t150) and 362 (t194) labelled nuclei and a second embryo (sequence 02, t150/t180). Nuclei measure $3.6\mu\text{m}$ across (median of the challenge’s segmentation masks); all spatial parameters below are fixed multiples of this diameter D .

Compressors. Luxar [2] (default settings, 1000 iterations, anisotropic voxel size, seeds $K \in \{10^3, \dots, 6.4 \times 10^4\}$; stored size = its .gsplats.zarr on disk). JPEG2000 (3D, OpenJPEG), JPEG-XL (per slice), ZFP fixed-rate, and trilinear downsampling (stored as 16-bit). Each codec’s quality parameter is bisected to the byte size of each Luxar fit; rows that cannot reach the target (JPEG-XL beyond $82\text{--}91\times$, ZFP below 1 bit/voxel) are flagged.

Detectors. (a) LoG: local maxima with a physically isotropic suppression window, ranked by scale-normalised LoG response at $\sigma \in \{0.2, 0.27, 0.33, 0.4\}D$; the top N are kept, with N equal to the number of labelled nuclei, where

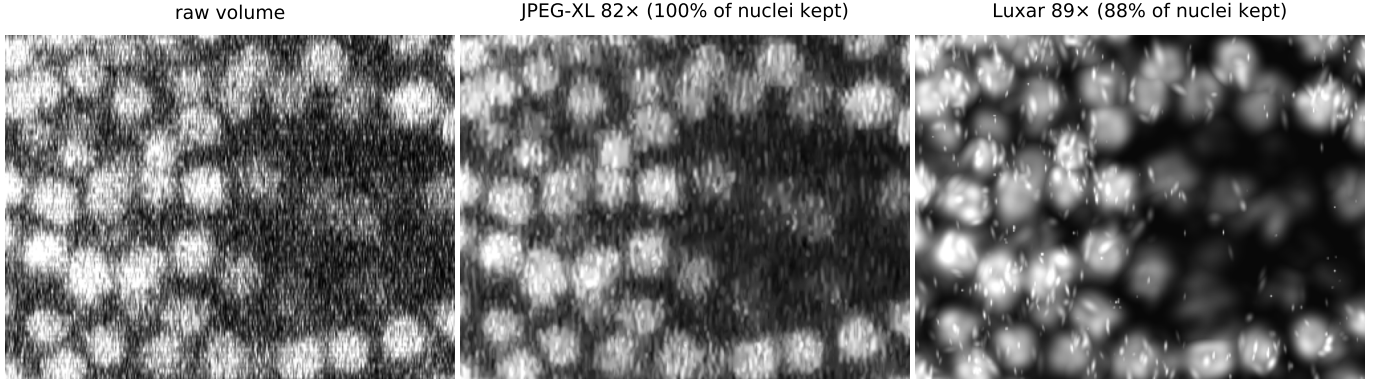


Fig. 1. The same region of *C. elegans* t194 (3-slice maximum projection) from the raw volume, JPEG-XL and Luxar at similar size. Titles give whole-frame nuclei kept by Cellpose. JPEG-XL preserves the textured appearance of nuclei; Luxar renders smooth blobs on a black background with thin streaks from elongated splats—an image distribution the learned segmenter was not trained on, while a blob detector is unaffected.

recall equals precision. (b) Cellpose 3 nuclei model [18], 2D per slice stitched across z , diameter D , scored at its own operating point.

Scoring. Detections are matched one-to-one to labels within $r = 0.6D$ by maximum cardinality, then minimum distance; results at $0.4D$ and $0.8D$ are in Table III. Nuclei kept = nuclei found in the reconstruction / found in the raw volume by the same detector (1.0 = nothing lost). Labelled nuclei are split into faint and bright halves by their LoG response in the raw volume. Survival is the label-free fraction of the raw volume’s 100 strongest LoG detections still found after compression. Every reconstruction and detection is saved; code and results are available [TODO: repository link].

IV. Results

Codecs. JPEG-XL keeps every nucleus up to its maximum compression under both detectors (Fig. 2). JPEG2000 is nearly lossless for detection to $\approx 60\times$ and then loses faint nuclei first: at $124\times$ on t194 it loses 37 faint but only 10 bright nuclei (LoG). Downsampling and fixed-rate ZFP fail early.

Gaussian splats. Beyond $120\times$, Luxar is the best option tested: at $\approx 250\times$ it keeps 98% (LoG) and 82% (Cellpose) of nuclei, against 78% and 74% for JPEG2000. Below $100\times$, its benefit depends on the detector. With the blob detector Luxar matches the codecs; with Cellpose it keeps 83–92% at $55\times$ to $94\times$, where JPEG-XL keeps 98–100%. The Cellpose deficit is flat across a $4\times$ range of splat budgets ($K=4k$ – $16k$), so it reflects the appearance of splat reconstructions—smooth, texture-free blobs with thin streak artefacts (Fig. 1) that a learned segmenter was not trained on—rather than a lack of capacity. The LoG detector responds most strongly to Gaussian blobs and thus flatters splat reconstructions; the codecs are insensitive to the detector choice.

The budget cliff. Loss is abrupt once splats become scarce relative to nuclei (Table II): at one splat per

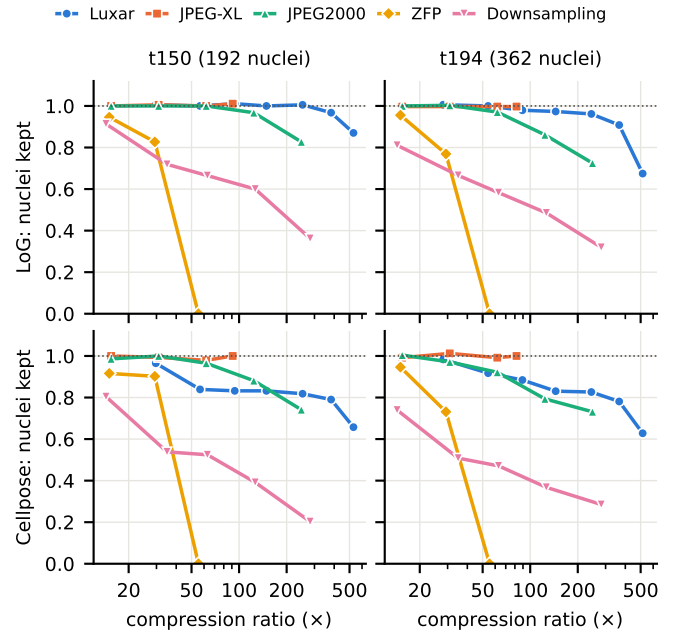


Fig. 2. Nuclei kept (relative to the raw volume) against compression ratio at matched bytes on disk, for a blob detector (top) and Cellpose (bottom), on two *C. elegans* frames. Luxar is blue. JPEG-XL cannot exceed 82–91 $\times$ on this data.

nucleus Luxar keeps 67% (LoG) / 63% (Cellpose) of nuclei, recovering to 96% / 83% at ≈ 7 per nucleus. Over the same budgets Luxar’s PSNR moves only within 19.1–21.3 dB (Fig. 3), and at ratios $\leq 100\times$ PSNR ranks Luxar below both codecs although the blob detector finds the same nuclei. Across all reconstructions, PSNR orders 7–12% of pairs separated by more than 1 dB in the wrong direction for faint nuclei kept.

A label-free guard. Survival, which needs no labels, tracks nuclei kept across Luxar budgets (Spearman $\rho = 0.68$, $n = 14$). Choosing the smallest budget with survival ≥ 0.95 keeps 97% (LoG) and 79–83% (Cellpose) of labelled

TABLE I

Nuclei kept (mean of two frames) near matched compression ratios, at match radius $0.6D$. **Sequence 02 to be added.**

Ratio	LoG			Cellpose		
	JXL	J2K	Luxar	JXL	J2K	Luxar
$\approx 30\times$	1.00	1.00	1.01	1.00	0.99	0.97
$\approx 60\times$	1.00	0.99	1.00	0.99	0.94	0.88
$\approx 90\times$	1.00	—	1.00	1.00	—	0.86
$\approx 135\times$	—	0.91	0.99	—	0.84	0.83
$\approx 250\times$	—	0.78	0.98	—	0.74	0.82

TABLE II

The budget cliff: Luxar fits by splats per labelled nucleus.

Frame	Splats	per nuc.	Ratio	PSNR	LoG	Cellpose	Surv.
t150	324	1.7	530 $\times$	19.6	0.87	0.66	0.92
t150	1,050	5.5	382 $\times$	19.9	0.97	0.79	1.00
t150	2,550	13.3	252 $\times$	20.0	1.01	0.82	0.99
t150	10,837	56.4	94 $\times$	20.3	1.01	0.83	1.00
t194	360	1.0	515 $\times$	19.1	0.67	0.63	0.74
t194	1,155	3.2	366 $\times$	19.6	0.91	0.78	0.87
t194	2,638	7.3	244 $\times$	19.8	0.96	0.83	0.90
t194	11,202	30.9	89 $\times$	20.1	0.98	0.88	0.96

nuclei: the check catches the budget cliff but, being itself a blob-detector measure, not the appearance gap of learned segmenters. **Luxar’s own self-calibrated budget K^* places at ... on this curve.**

V. Discussion and Limitations

Gaussian splats compress fluorescence volumes further than standard codecs while keeping most nuclei detectable—but only above a budget that PSNR cannot identify, and with a detector-dependent loss for learned segmenters that no image-fidelity metric exposes. We therefore recommend (i) JPEG-XL wherever its ratio suffices; (ii) for higher ratios, validating splat budgets with an object-level check such as survival, and (iii) reporting rate–detectability curves with the detector that will be used downstream.

Limitations: one dataset family (**two embryos**); Cellpose is used in its 2D-stitched mode; Luxar fitting takes 17–46 min per volume on one A100 GPU versus seconds for the codecs; the faint/bright split is defined by one detector’s response; and we test a single Gaussian compressor with default settings. Retraining or fine-tuning segmenters on splat reconstructions may close the appearance gap and is future work, as is the same audit for small low-contrast lesions in CT.

VI. Conclusion

Measured at matched bytes against human labels, Gaussian-splat compression is the strongest option for nucleus detection at high ratios, yet its safety depends on the budget and on the downstream detector—two things

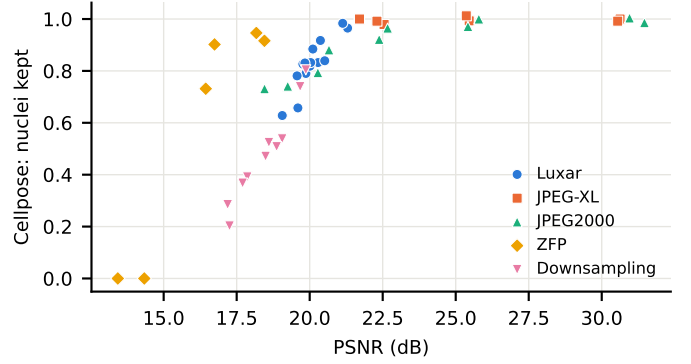


Fig. 3. PSNR against nuclei kept (Cellpose) for every reconstruction. Luxar’s PSNR barely changes while nuclei kept ranges from 63% to near 1.

TABLE III

Match-radius sensitivity: nuclei found in the raw volume and nuclei kept by Luxar at $\approx 90\times$ (t150).

Radius	LoG			Cellpose		
	raw	JXL	Luxar	raw	JXL	Luxar
$0.4D$	179	1.02	1.03	115	1.02	0.76
$0.6D$	185	1.01	1.01	143	1.00	0.83
$0.8D$	186	1.01	1.01	146	1.01	0.87

PSNR does not show. Object-level rate–detectability evaluation should accompany Gaussian volume compression.

References

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3D gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, 2023.
- [2] L. A. Royer, “Luxar: Gaussian splatting for microscopy and scalable interactive web visualisation of multidimensional scientific data,” *Zenodo*, <https://zenodo.org/records/22912050>, 2026, tODO verify author list; software at <https://github.com/royerlab/luxar>.
- [3] tODO verify authors, “GaussianPile: A unified sparse Gaussian splatting framework for slice-based volumetric reconstruction,” *arXiv:2603.20611*, 2026.
- [4] —, “GSToken: Geometry-structured Gaussian tokens for compact 3D medical image representation,” *arXiv:2608.17425*, 2026.
- [5] M. R. Martin et al., “Fixed-budget Gaussian volume encoding with structure-aware allocation,” *arXiv:2608.14112*, 2026, tODO verify author list.
- [6] Y. Li et al., “3DGR-CT: Sparse-view CT reconstruction with a 3D Gaussian representation,” *Medical Image Analysis*, 2025, tODO verify author list, volume and pages.
- [7] R. Zha, T. J. Lin, Y. Cai, J. Cao, Y. Zhang, and H. Li, “R²-Gaussian: Rectifying radiative Gaussian splatting for tomographic reconstruction,” in *Advances in Neural Information Processing Systems*, 2024.
- [8] J. Batson and L. Royer, “Noise2Self: Blind denoising by self-supervision,” in *Proceedings of the 36th International Conference on Machine Learning*, 2019, pp. 524–533.
- [9] D. A. Koff and H. Shulman, “An overview of digital compression of medical images: can we use lossy image compression in radiology?” *Canadian Association of Radiologists Journal*, vol. 57, no. 4, pp. 211–217, 2006.
- [10] F. Amat, W. Lemon, D. P. Mossing, K. McDole, Y. Wan, K. Branson, E. W. Myers, and P. J. Keller, “Fast, accurate reconstruction of cell lineages from large-scale fluorescence

- microscopy data,” *Nature Methods*, vol. 11, no. 9, pp. 951–958, 2014.
- [11] M. Dannecker, S. Jia, N. Stolt-Ansó, N. Girard, G. Auzias, F. Rousseau, and D. Rueckert, “Fast and explicit: Slice-to-volume reconstruction via 3D Gaussian primitives with analytic point spread function modeling,” in *Medical Imaging with Deep Learning (MIDL)*, 2026, tODO verify author first names.
 - [12] D. S. Taubman and M. W. Marcellin, *JPEG2000: Image Compression Fundamentals, Standards and Practice*. Kluwer Academic Publishers, 2002.
 - [13] J. Alakuijala et al., “JPEG XL next-generation image compression architecture and coding tools,” in *Applications of Digital Image Processing XLII*, Proc. SPIE 11137, 2019.
 - [14] P. Lindstrom, “Fixed-rate compressed floating-point arrays,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 20, no. 12, pp. 2674–2683, 2014.
 - [15] V. Ulman et al., “An objective comparison of cell-tracking algorithms,” *Nature Methods*, vol. 14, pp. 1141–1152, 2017.
 - [16] M. Maška et al., “The cell tracking challenge: 10 years of objective benchmarking,” *Nature Methods*, vol. 20, pp. 1010–1020, 2023.
 - [17] T. Lindeberg, “Feature detection with automatic scale selection,” *International Journal of Computer Vision*, vol. 30, no. 2, pp. 79–116, 1998.
 - [18] C. Stringer, T. Wang, M. Michaelos, and M. Pachitariu, “Cellpose: a generalist algorithm for cellular segmentation,” *Nature Methods*, vol. 18, pp. 100–106, 2021.